Question	Answer	Marks
1	**Defining the problem**	
	z is the independent variable and t is the dependent variable **or** vary z and measure t	**1**
	keep B **and** A constant	**1**
	Methods of data collection	
	labelled diagram of workable experiment including: • pin/rod though hole • supported by a stand • sheet able to oscillate freely • at least one label from copper/sheet, hole, clamp stand, rod, pin	**1**
	use of stop-watch/timer to measure t (from release to stopping) **or** use of stop-watch/timer to measure time for the sheet (to stop) oscillating	**1**
	use of micrometer to measure z	**1**
	use of rule(r) to measure lengths to determine A **and** A = length × breadth	**1**
	Method of Analysis	
	plot a graph of lg t against lg z or equivalent (e.g. ln t against ln z)	**1**
	q = gradient	**1**
	$K = AB\rho \times 10^{y\text{-intercept}}$ ($K = AB\rho \times e^{y\text{-intercept}}$ for ln t against ln z)	**1**

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Question	Answer	Marks
1	**Additional detail including safety considerations**	**6**
	D1 use of cushion / sand box <u>in case sheet falls</u> **or** use gloves to protect hands from <u>cuts / sharp edges</u>	
	D2 keep (initial) distance between (copper) sheet and (poles of) magnet <u>constant</u> **or** keep (initial) distance between (copper) sheet and coil(s) <u>constant</u>	
	D3 keep initial displacement (of copper sheet) <u>constant</u>	
	D4 method to ensure initial displacement (of copper sheet) is constant, e.g. initially line up (corner of) plate with fiducial marker/vertical pin	
	D5 relationship valid <u>if</u> a straight line (with y-intercept = $\log\left(\frac{K}{AB\rho}\right)$) is produced	
	D6 repeat measurements of z <u>in different positions</u> and <u>average</u> z	
	D7 measure B / magnetic flux density using a (calibrated) Hall probe	
	D8 additional detail on use of Hall probe, e.g. adjust (position of) probe until maximum value **or** measure B using Hall probe first in one direction and then in the opposite direction and average	
	D9 drawn method to create a magnetic field perpendicular to the area of the sheet, e.g. pair of magnets/horseshoe magnet/pair of coils connected to a (d.c.) supply	
	D10 repeat experiment for each z and average t	
	D11 method to determine ρ, e.g. measure mass with balance **and** volume = Az **and** density = mass / volume	

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